

BANK ATM MANAGER



Name:

Muhammad Abu Bakar

Roll Number:

BSCS25109

Degree:

BS Computer Science (Semester I)

Section:

A

Subject:

Programming Fundamental

PROJECT:

01

```

int main()
{
    loadAccounts();

    int ch;
    do
    {
        Sleep(1500);
        system("cls");
        cout << "<=====>" << endl;
        cout << "          BANK ATM SYSTEM          " << endl;
        cout << "<=====>" << endl;
        cout << "1. Customer Login" << endl;
        cout << "2. Manager Login" << endl;
        cout << "3. Exit" << endl << endl;
        cout << "<=====>" << endl << endl;
        cout << "Enter choice: ";
        cin >> ch;

        if (ch == 1)
        {
            customerLogin();
        }
        else if (ch == 2)
        {
            if (managerLogin() == 1)
            {
                managerMenu();
            }
        }

    } while (ch != 3);
}

```

```

void customerLogin()
{
    long long account;
    int index = 0;
    cout << "Enter account number: ";
    cin >> account;

    for (int i = 0; i < totalAccounts; i++)
    {
        if (accountNumbers[i] == account)
        {
            index = i;
        }
    }

    if (index == -1)
    {
        cout << "Account not found"<<endl;
        return;
    }
    if (status[index] != 'A')
    {
        cout << "Account not active"<<endl;
        return;
    }
    if (locked[index])
    {
        cout << "Account locked"<<endl;
        return;
    }

    int attempts = 0;
    while (attempts < loginAttempts)
    {
        cout << "Enter PIN: ";

        char pin[6];
        char ch;
        int i = 0;
        while ((ch = _getch()) != 13 && i < 5)

```

```

int found = 1;
for (int j = 0; j < 6; j++)
{
    if (pins[index][j] != pin[j])
    {
        found = 0;
        break;
    }
    if (pins[index][j] == '\0' && pin[j] == '\0')
    {
        break;
    }
}

if (found)
{
    failedAttempts[index] = 0;
    saveAccounts();

    if (needsApproval[index])
    {
        cout << "Login successful but your account is pending admin approval"<<endl;
        cout << "You cannot perform any transactions until approved"<<endl;
        return;
    }

    cout << "Login successful"<<endl;

    customerMenu(index);
    return;
}
else
{
    attempts++;
    failedAttempts[index]++;
    cout << "Incorrect PIN. Attempts left: " << (3 - attempts) << endl;
}
}

locked[index] = 1;
saveAccounts();

```

```

void customerMenu(int index)
{
    int c;
    do {
        system("cls");
        cout << "<=====>" << endl;
        cout << "          CUSTOMER MENU          " << endl;
        cout << "<=====>" << endl;
        cout << "1. Deposit" << endl;
        cout << "2. Withdraw" << endl;
        cout << "3. Check Balance" << endl;
        cout << "4. Transfer" << endl;
        cout << "5. Change PIN" << endl;
        cout << "6. Mini Statement" << endl;
        cout << "7. Logout" << endl<<endl;
        cout << "<----->" << endl;
        cout << "Enter choice: ";
        cin >> c;
    }
}

```

```
    if (c == 1)
    {
        deposit(index);
    }
    else if (c == 2)
    {
        withdraw(index);
    }
    else if (c == 3)
    {
        checkBalance(index);
    }
    else if (c == 4)
    {
        transfer(index);
    }
    else if (c == 5)
    {
        changePIN(index);
    }
    else if (c == 6)
    {
        miniStatement(accountNumbers[index]);
    }

    if (c != 7)
    {
        cout <<endl<< "Press any key to continue";
        getch();
    }
} while (c != 7);
```

```

void managerMenu()
{
    int ch;
    do
    {
        system("cls");
        cout << "=====>" << endl;
        cout << "          MANAGER MENU          " << endl;
        cout << "=====>" << endl;
        cout << "1. Open New Account" << endl;
        cout << "2. Close Account" << endl;
        cout << "3. Deactivate/Activate" << endl;
        cout << "4. Unlock Account" << endl;
        cout << "5. Search Account" << endl;
        cout << "6. Display All" << endl;
        cout << "7. Summary Report" << endl;
        cout << "8. Approve Accounts" << endl;
        cout << "9. Back" << endl<<endl;
        cout << "=====>" << endl<<endl;
        cout << "Enter choice: ";
        cin >> ch;
    }
}

```

```
    if (ch == 1)
    {
        openAccount();
    }
    else if (ch == 2)
    {
        closeAccount();
    }
    else if (ch == 3)
    {
        changeAccountStatus();
    }
    else if (ch == 4)
    {
        unlockAccount();
    }
    else if (ch == 5)
    {
        searchAccount();
    }
    else if (ch == 6)
    {
        displayAllAccounts();
    }
    else if (ch == 7)
    {
        summaryReport();
    }
    else if (ch == 8)
    {
        approveAccounts();
    }

    if (ch != 9)
    {
        cout <<endl<< "Press any key to continue";
        getch();
    }
} while (ch != 9);
```



```

int managerLogin()
{
    ifstream fin("admin.txt");

    if (!fin)
    {
        cout << "File not found" << endl;
        return 0;
    }

    char encryptedPass[40];
    char realPass[40];
    char inputPass[40];
    fin >> encryptedPass;
    fin.close();

    int i = 0;
    while (encryptedPass[i] != '\0')
    {
        realPass[i] = encryptedPass[i] - 5;
        i++;
    }
    realPass[i] = '\0';

    cout << "Enter Manager Password: ";
    char ch;
    int j = 0;
    while ((ch = _getch()) != 13 && j < 39)
    {
        inputPass[j] = ch;
        cout << '*';
        j++;
    }
    inputPass[j] = '\0';
    cout << endl;

    int match = 1;
    for (int j = 0; realPass[j] != '\0' || inputPass[j] != '\0'; j++)
    {
        if (realPass[j] != inputPass[j])
        {
            match = 0;
            break;
        }
    }
}

```

```

if (match)
{
    cout << "Manager login successful" << endl;
    return 1;
}
else
{
    cout << "Invalid manager password please try again " << endl;
    return 0;
}

```

```
system("cls");
cout << "<=====>" << endl;
cout << "    Thank you for using BANK ATM SYSTEM    " << endl;
cout << "          Your money is safe                    " << endl;
cout << "                /\_/_/\                        " << endl;
cout << "                ( o.o )                        " << endl;
cout << "                > ^ <                          " << endl;
cout << "          Have a secure and pleasant day        " << endl;
cout << "<=====>" << endl;
```



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<=====>

BANK ATM SYSTEM

<=====>

1. Customer Login
2. Manager Login
3. Exit

<=====>

Enter choice: |

<=====>

BANK ATM SYSTEM

<=====>

1. Customer Login
2. Manager Login
3. Exit

<=====>

Enter choice: 1

Enter account number: 10001

Enter PIN: *****

Incorrect PIN. Attempts left: 2

Enter PIN: **

Incorrect PIN. Attempts left: 1

Enter PIN: *|

<=====>

MANAGER MENU

<=====>

1. Open New Account
2. Close Account
3. Deactivate/Activate
4. Unlock Account
5. Search Account
6. Display All
7. Summary Report
8. Approve Accounts
9. Back

<=====>

Enter choice: 4

Enter Account Number to unlock: 10001

Account unlocked

Press any key to continue|

<=====>

MANAGER MENU

<=====>

1. Open New Account
2. Close Account
3. Deactivate/Activate
4. Unlock Account
5. Search Account
6. Display All
7. Summary Report
8. Approve Accounts
9. Back

<=====>

Enter choice: 1

New Account Number assigned: 10007

Enter 4-digit PIN: ***

PIN must be exactly 4 digits, Try again

Enter 4-digit PIN: **

PIN must be exactly 4 digits, Try again

Enter 4-digit PIN: ***

PIN must be exactly 4 digits, Try again

Enter 4-digit PIN: ****

Re-enter PIN: ****

Enter initial deposit : 111111

Account created successfully

New Account Number is: 10007

Press any key to continue|

<=====>

MANAGER MENU

<=====>

1. Open New Account
2. Close Account
3. Deactivate/Activate
4. Unlock Account
5. Search Account
6. Display All
7. Summary Report
8. Approve Accounts
9. Back

<=====>

Enter choice: 8

<=====>

Accounts Pending Approval

<=====>

Account: 10007

Balance: 111111

Enter account number to approve : |

<=====>

MANAGER MENU

<=====>

1. Open New Account
2. Close Account
3. Deactivate/Activate
4. Unlock Account
5. Search Account
6. Display All
7. Summary Report
8. Approve Accounts
9. Back

<=====>

Enter choice: 7

<=====>

Summary Report

<=====>

1. Active: 7
2. Closed: 0
3. Deactivated: 0
- 4 .Locked: 2
5. Total Balance: 592111

Press any key to continue|

```
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<=====>
      BANK ATM SYSTEM
<=====>
1. Customer Login
2. Manager Login
3. Exit

<=====>

Enter choice: 1
Enter account number: 10001
Account locked
|
```

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